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(50)

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End Semester Examination 2024

Name of the Course: B. Tech (Civil Engineering)
Name of the Paper: Hydraulics and Hydraulic Machines
Time: 3 Hours

Semester: IV
Paper Code: TCE 401
Maximum Marks: 100

Note:
All questions are compulsory
Answer any two sub questions among a, b & c in each main questions
Total marks in each question are twenty
Each question carries 10 marks

(20 marks)

Q1

(a) Derive Manning's equation for Uniform Flow in an open channel. What are the dimensions of the constant n in Manning's formula?

(b) Determine the most economical cross section o a rectangular channel which is to be dug in the rocky portion o a soil. The channel is to convey $8m^3/s$ of water with an average velocity o 2 m/s. The value of Chezy's C = 65. **CO1**

(c) The specific Energy for a 3 m wide channel is to be 3Nm/N. What would be the maximum possible discharge?

(20 marks)

Q2

(a) Derive the dynamic equation of Gradually Varied Flow (GVF) in an open channel.

(b) A hydraulic jump occurs in a 0.5 m wide rectangular channel at the point where the depth of flow is 0.15 m and the Froude No. is 2.5. Make calculations for the specific energy, critical and sequent depths and energy loss due to hydraulic jump. **CO2**

(c) A river flowing towards a sea at a velocity of 5 km/Hour and having a depth of 2 m encounters an incoming tide which increases the height to 5 m and a tidal bore travels upstream. Estimate the speed of the bore and speed and direction of flow after the passage of the bore.

(20 marks)

Q3

(a) With the help of neat sketch, explain the circulation theory of lift and Magnus Effect.

(b) The wings of a model aeroplane are tested in a wind tunnel under atmospheric pressure and a wind velocity of 2.50 m/s. The aerofoil wing may be taken as equivalent to a flat plate of length 180 mm and width 600 mm. Determine the boundary layer thickness at trailing edge and the drag force exerted by the wind. Take for air, density = $1.22\ kg/m^3$ and kinematic viscosity = $1.5 \times 10^{-4}\ m^2/s$. **CO3**

(c) An airplane weighing 117720 N has a wing span of 12 m and a wing area of $20\ m^2$. It flies at a velocity of 360 km/hour at a steady level in still air. Find the lift coefficient, the total drag on the wing taking $C_d = 0.06$ and the power required to keep the airplane at this velocity.

Q4 **(20 marks)**

(a) Explain with sketches the working principle of a single stage centrifugal pump.

(b) A 65 mm diameter jet moving at 25 m/s impinges on a series of flat vanes mounted on a large wheel. If the velocity of vane is 6.5 m/s, find the force exerted by the jet, work done per second and the hydraulic efficiency. **CO4**

(c) A centrifugal pump running at 1450 rpm discharges 110 liters per second against a head of 23 m. If the diameter of the impeller is 250 mm and its width is 50 mm, find the vane angle at the outer periphery. The manometric efficiency of the pump is 75%.

Q5 **(20 marks)**

(a) Define and explain hydraulic efficiency, mechanical efficiency and overall efficiency of a turbine.

(b) Find how many jets are required for a Pelton wheel to develop a shaft power of 8975 kW under a head of 265 m at a speed of 500 rpm. The jet is not to exceed 1/9 of the wheel diameter. Find also the diameters of the jet, the wheel and the quantity of water needed assuming an oveall efficiency of 87%. Take Cv = 0.98 and peripheral velocity = 0.45 $\sqrt{(2gh)}$. **CO5**

(c) An inward flow reaction turbine running at 500 rpm has an external diameter of 700 mm and a width of 180 mm. If the guide vanes are at 20^0 to the wheel tangent and the absolute velocity of water at inlet is 25 m/s, find the discharge of the turbine and the runner vane angle at inlet.

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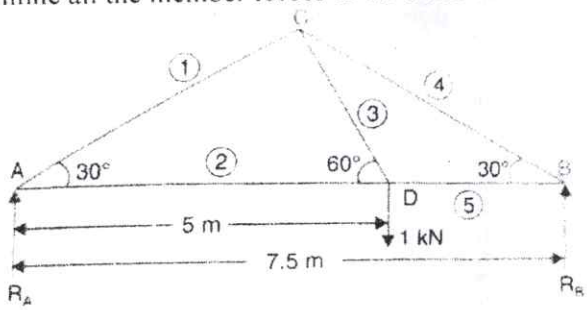
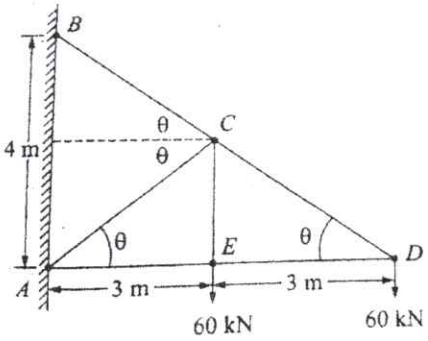
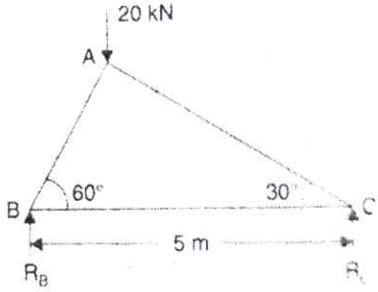
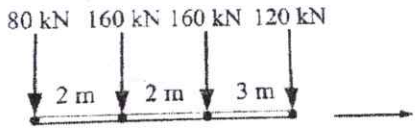
EVEN END SEMESTER EXAMINATION- 2024

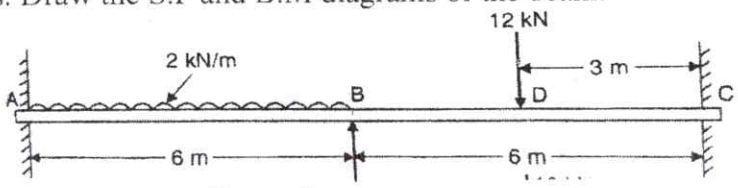
Name of the Course: B.Tech
Name of the Paper : Structural Analysis- I
Time: 3 Hours

Semester : IV semester
Paper Code: TCE 402
Maximum Marks: 100

Note: i) All questions are compulsory
ii) Answer any two sub questions among a, b & c in each main question
iii) Total marks in each main question are twenty
iv) Each question carries 10 marks.

Q.1		(20 Marks)	
	a)	Determine all the member forces in the truss shown in figure 1 by the method of joints.  Figure 1	(CO 1)
	b)	Determine the forces in the members of the truss as shown in figure 2.  Figure 2	(CO 1)
	c)	Determine all the member forces in the truss shown in figure 3 by the method of Tension Coefficients.  Figure 3	(CO 1)
Q.2		(20 Marks)	
	a)	A simply supported beam AB of span 4m carries a point load of 100KN at its center C. The values of I for the left half is 1×10^8 mm^4 and for the right half portion I is 2×10^8 mm^4. Find the slopes at the two supports and deflection under the load using conjugate beam method. Take $E = 2 \times 10^5$ N/mm^2	(CO 2)

	b)	A cantilever of length 2m carries a uniformly distributed load 2kN/m over a length of 1m from the free end, and a point load of 1kN at the free end. Find the slope and deflection at the free end if $E=2\times10^5$ N/mm^2 and $I= 6.667\times10^7$ mm^4	(CO 2)
	c)	The external diameter of a hollow shaft is twice the internal diameter. It is subjected to pure torque and it attains a maximum shear stress τ. Show that the strain energy stored per unit volume of the shaft is $\frac{5\tau^2}{16C}$. Such a shaft is required to transmit 5400kW at 110 r.p.m with uniform torque, the maximum stress not exceeding 84 MN/m^2. Determine: i) The shaft diameters ii) The energy stored per m^3	(CO 2)
Q.3		(20 Marks)	
	a)	A simply supported beam has a span of 15m. Uniformly distributed load of 40kN/m and 5m long crosses the girder from left to right. Draw the influence line diagram for shear force and bending moment at a section 6 m from left end. Use these diagrams to calculate the maximum shear force and bending moment at this section.	(CO 3)
	b)	For point loads, 8,15.15 and 10 kN have centre spacing of 2 m between consecutive loads and they traverse a girder of 30 m span from left to right with 10kN load lending. Calculate the maximum bending moment and shear force at 8 m from the left support	(CO 3)
	c)	A train of wheel loads as shown in figure 4 crosses a girder of 25m span with 120 kN load leading. Determine the value of i) Maximum bending moment at the section 8m from the left end of the girder ii) Absolute maximum bending moment of the girder  Figure 4	(CO 3)
Q.4		(20 Marks)	
	a)	A three hinged circular arch hinged at the springing and crown points has a span of 40m and a central rise of 8m. It carries a uniformly distributed load 20kN/m over the left half of the span together with a concentrated load of 100kN at the right quarter span point. Find the reactions at the supports, normal thrust and shear at a section 10m from left support.	(CO 4)
	b)	A circular arch to span 25m with a central rise 5m is hinged at the crown and springing. It carries a point load of 100kN at 6m from the left support. Calculate i) The reactions at the supports ii) The reactions at crown iii) Moment at 5m from the left support	(CO 4)
	c)	A three hinged parabolic arch hinged at the supports and at the crown has a span of 24m and a central rise of 4m. It carries a concentrated load of 50kN at 18m from left support and a uniformly distributed load of 30kN/m over the left-half portion. Determine the moment, thrust and radial shear at a section 6m from the left support.	(CO 4)
Q.5		(20 Marks)	
	a)	A continuous beam ABC covers two consecutive span AB and BC of lengths 4m and 6m, carrying uniformly distributed loads of 6kN/m and 10kN/m respectively. If the ends A and C are simply supported, find the support moments A, B and C. Draw also B.M. and S.F. diagrams.	(CO 5)

	b)	A continuous beam ABC of uniform section, with span AB and BC as 4m each, is fixed at A and simply supported at B and C. The beam is carrying a uniformly distributed load of 6kN/m run throughout its length. Find the support moments and the reactions. Also draw the bending moment and shear force diagrams.	(CO 5)
	c)	A continuous beam ABC of uniform section, with span AB and BC as 6m each, is fixed at A and C and supported at B as shown in figure 5. Find the support moments and reactions. Draw the S.F and B.M diagrams of the beam.  Figure 5	(CO 5)

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EVEN END SEMESTER EXAMINATION - 2024

Name of the Course: B. Tech Civil Engineering
Name of the Paper: Advanced Surveying
Time: 3 hours

Semester: IV
Paper Code: TCE 403
Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20 marks)	**CO1**
(a)	Differentiate between primary, secondary, and tertiary triangulation systems.	
(b)	Define the following terms: i) True value of quantity ii) Most probable value iii) True Error iv) Most probable error v) Residual error	
(c)	What are the three kinds of errors in measurement? Explain consequences for each of them if undetected.	
Q2	(20marks)	**CO2**
(a)	Find the normal equations of x, y and z from the given set of indirectly observed quantities of equal weights $3x+3y+z-4=0$ $x+2y+2z-6=0$ $5x+y+4z-21=0$	
(b)	Find most probable values of angles A, B, C of ΔABC from the following observed equations using method of correlates. A= 77°14'20" having w=4 B= 49°40'35" having w=3 C= 53°04'52" having w=2	
(c)	Find most probable value of angle A: i) A= 30°28'40"; 3A= 90°25'55"; 4A= 121°54'30" ii) A= 30°28'40" having w=2; 3A= 91°25'55" having w= 3	
Q3	(20marks)	**CO3**
(a)	Define and give expressions of all the elements of simple circular curve and draw a well labeled diagram.	
(b)	Calculate the ordinates at 10 m distances for a circular curve having a long chord of 100 meters and a mid-ordinate of 5 m.	
(c)	What are various methods of setting out simple circular curves? Explain the method of setting out curve by method of ordinates or offsets from long chord. Support your answer with diagram and expressions.	
Q4	(20marks)	**CO4**
(a)	Define the following terms: 1. Spherical triangle 2. Astronomical triangle 3. Nadir 4. Zenith	

	5. Azimuth	
(b)	Differentiate between a map and an aerial photograph. Define the following terms with a diagram: 1. Tilted photograph 2. Oblique photograph 3. Exposure station 4. Flying height 5. Iso centre	
(c)	Two points A and B having elevations of 650 m and 250 m, respectively, above datum, appear on a vertical photograph obtained with a camera of focal length 250 mm and flying height of 2700 m above datum. Their correlated photographic coordinates are as follows:	

Point	Photographic Coordinates	
	x (cm)	y (cm)
a	+3.65	+2.54
b	-2.25	+5.59

Determine the length of line AB.

Q5	(20marks)	
(a)	A line 2350 m long lying at an elevation of 500 m measures 10.50 cm on a vertical photograph. The focal length of the camera used is 20 cm. Determine the scale of the photograph for an area having an elevation of 1200 m.	**CO5**
(b)	Write short notes on the following: 1. NCC 2. Sun-synchronous satellites 3. GIS 4. Remote Sensing 5.Passive sensors	**CO6**
(c)	Give a detailed overview on GPS. What is the principle of GPS? Explain the three segments of GPS?	**CO6**

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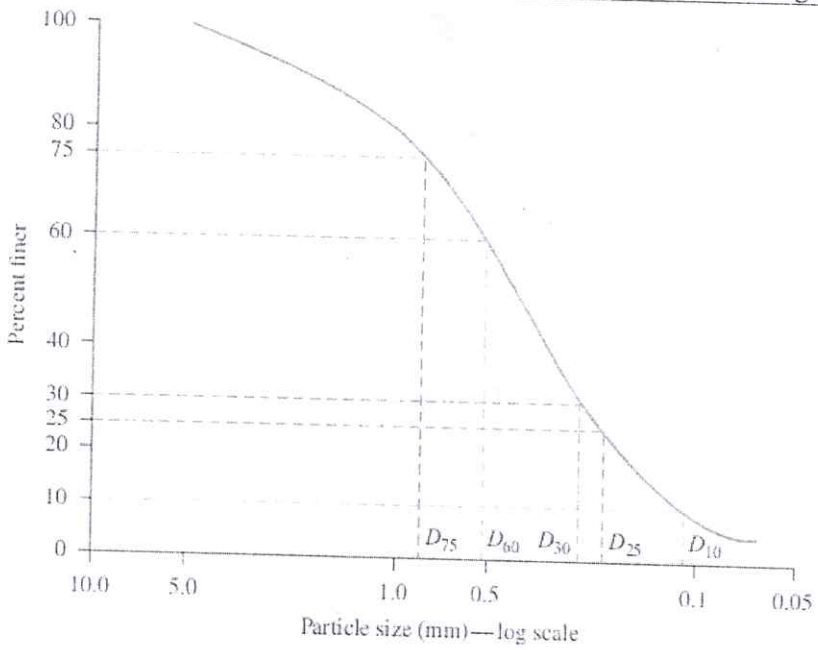
End Semester Examination 2024

Name of the Course: B. Tech
Name of the Course: Geotechnical Engg.-I
Time: 3 hours

Semester: IV
Course Code: TCE 405
Maximum Marks: 100

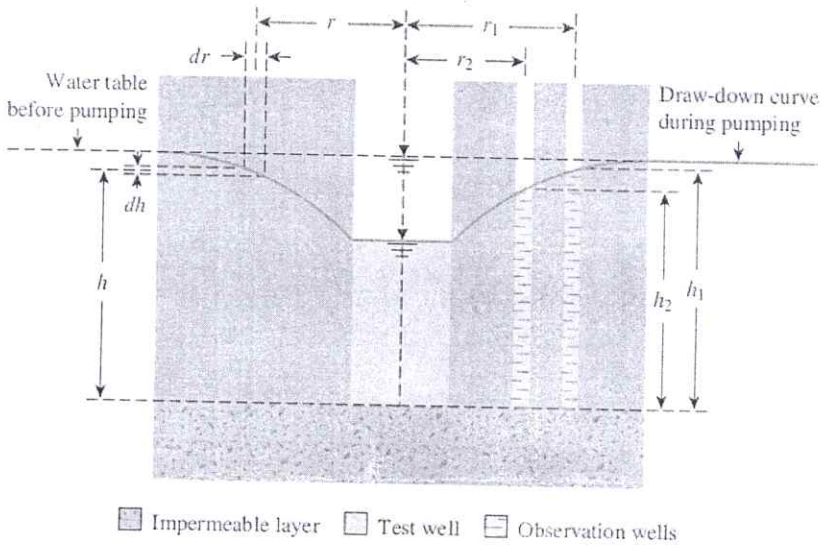
Note:-

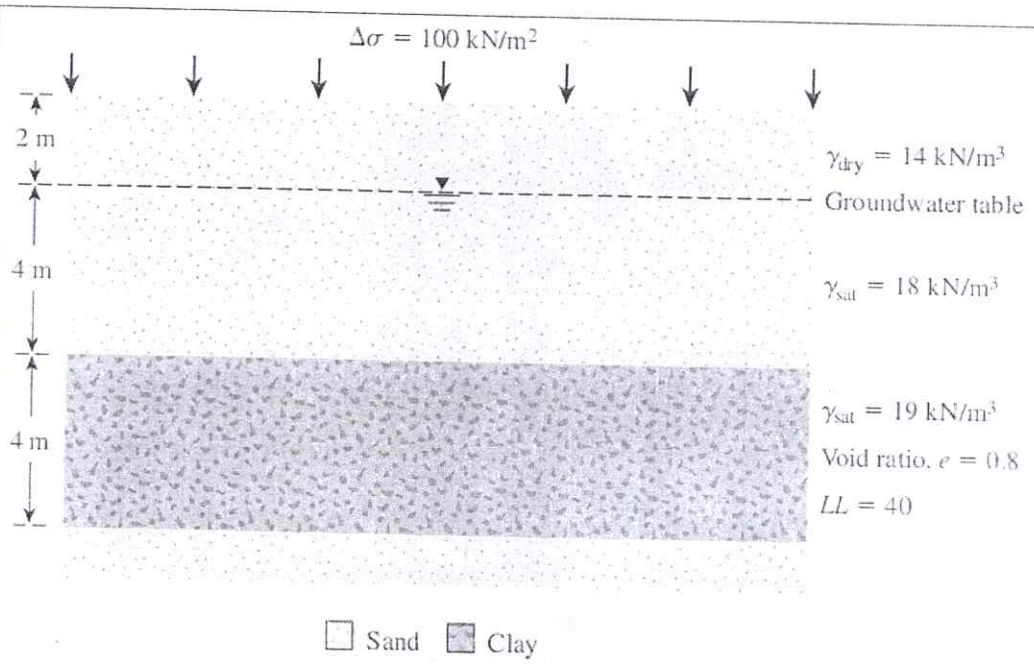
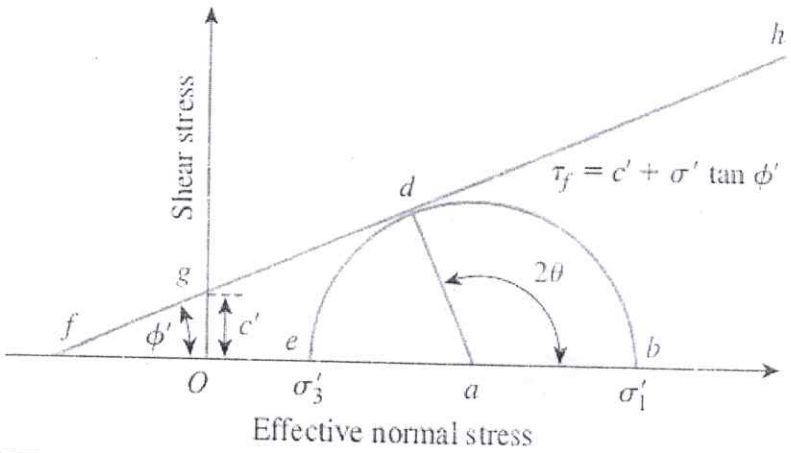
(i) Answer all the 5 questions by choosing **any two** of the sub questions.
(ii) Each sub-question carries 10 marks

Q1	**(20marks)**	
(a)	Define the subject soil mechanics. How it is different from Geotechnical Engineering? Discuss the importance of the subject in the field of civil engineering and also its future directions.	
(b)	Discuss the major soil deposit in India and its aerial distribution in Indian Subcontinent. How black cotton soil is different from the soil of Indo-Gangetic plain	
(c)	 Calculate the coefficient of uniformity and coefficient of curvature from the grain size distribution of the coarse grained material shown in the figure. Also, classify the material as per Indian Standard Code of practice.	CO1
Q2	**(20 marks)**	
(a)	Differentiate Relative Density and Relative Compaction. For a given soil, $e_{max} = 0.75$ and $e_{min} = 0.4$. Let's take $G_s = 2.68$. In the field, the soil is compacted to a density of 18 kN/m^3 at a moisture content of 12%. Determine the relative density value of the compacted soil. If the same soil was tested in the lab and MDD was obtained as 21 kN/m^3, also calculate the relative compaction value.	CO2
(b)	Following are the results of a shrinkage limit test: • Initial volume of soil in a saturated state = 24.6 cm^3 • Final volume of soil in a dry state = 15.9 cm^3 • Initial mass in a saturated state = 44.0 gm • Final mass in a dry state = 30.1 gm Determine the shrinkage limit of the soil.	

(c)	Discuss the procedure to classify the fine grained soil as per Indian Standard code of practice using plasticity chart.	
Q3	**(20 marks)**	CO3
(a)	What do you understand by the term "Light Compaction" and "Heavy Compaction" tests performed in the lab? Compare the input energy (E) given to soil by the two methods of compaction and their ratio , i.e. E_{heavy} / E_{light}	
(b)	Discuss the step by step procedure to find the field density of soil using sand replacement method.	
(c)	Laboratory compaction test results for a clayey silt are given in the following table. Use normal graph paper and obtain the MDD and OMC values. Also, draw the ZAV line on the same plot.	

Moisture content (%)	Dry unit weight (kN/m³)
6	14.80
8	17.45
9	18.52
11	18.9
12	18.5
14	16.9

Q4	**(20 marks)**	CO4
(a)	 Develop the expression for the field coefficient of permeability from the Pumping test from a well in an unconfined permeable layer underlain by an impermeable stratum	

(b)	 A soil profile is shown in Figure. If a uniformly distributed load, $\Delta\sigma$, is applied at the ground surface, what is the settlement of the clay layer caused by primary consolidation if the clay is normally consolidated.	
(c)	Discuss the following i) Coefficient of consolidation ii) Coefficient of compression iii) Pre-consolidation pressure iv) Over consolidation ratio	
Q5	**(20 marks)**	
(a)	Derive the expression for Mohr-Coulomb Failure hypothesis with the help of following diagram: 	**CO5**
(b)	Discuss the following i) Hook's Law ii) Isobar or pressure bulb iii) Relation between elastic modulus and shear modulus iv) Linear and non-linear Stress-strain curve	

(c) Following table provides the results of direct shear test conducted on a cohesive-frictional soil sample. If the area of the specimen was 0.0019634 m^2, estimate the peak strength and residual strength of the soil.

Test no.	Normal force, N (N)	Shear force at failure, S_{peak} (N)	Residual shear force, $S_{residual}$ (N)
1	150	157.5	44.2
2	250	199.9	56.6
3	350	257.6	102.9
4	550	363.4	144.5

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End Semester Examination June 2024

Name of the Course: B. Tech
Name of the Paper: Engineering Geology
Time: 3:00 Hours

Semester: IV
Course Code: TCE 406
Maximum Marks: 100 .

Note: -

(i) **All questions are compulsory.**
(ii) **Answer any two sub questions among a, b & c in each main question.**
(iii) **Total marks in each question are twenty.**
(iv) **Each question carries 10 marks**

Q1 **(20marks)**

(a) Write a short note on the following group of minerals:
i) Feldspar group
ii) Silicate group
iii) Pyroxene group — CO1

(b) What are rock minerals and discuss any 5 physical properties of a mineral.

(c) Discuss in detail about the importance of geology in civil engineering.

Q2 **(20 marks)**

(a) What are concordant and discordant bodies and briefly explain various concordant and discordant bodies. — CO2

(b) What are metamorphic rocks? Classify metamorphic rocks?

(c) Discuss the engineering properties of rocks. Also mention five examples each of igneous, sedimentary, and metamorphic rocks.

Q3 **(20 marks)**

(a) What are folds and folding? Discuss various types of folds with the help of neat diagram. — CO3

(b) Enlist some internal and external factors which causes landslides. Classify various types of landslide.

(c) Discuss about the causes and classification of various types of joints in rocks with the help of neat diagram.

Q4 **(20 marks)**

(a) Explain the following terminologies related to earthquake: — CO4
i) Focus
ii) Epicenter
iii) Magnitude
iv) Seismograph
v) Seismology

(b) What is underground water and what are the different sources of underground water?

(c) Discuss the zonal classification for seismic zones in India.

Q5 **(20 marks)**

(a) Define tunnel and discuss about geological investigation for site selection of tunnel? **CO5**

(b) With the help of neat diagram kindly discuss various methods of geo-physical explorations along with their working principles.

(c) Define reservoir and discuss about geological investigation for site selection of reservoir?